AI assisted Coding lab(3.2)

Name : Neerudu Gifty Taniksha
Roll Number : 2503A51107
Batch : 38

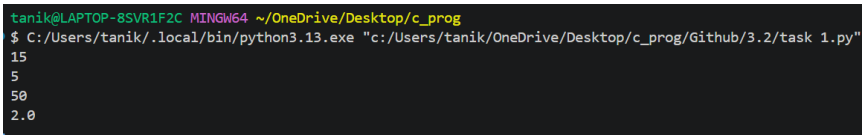
Task 1

Prompt and Code

```
# Create a Python function named calculator().
# The calculator should perform basic arithmetic operation
s:
# addition, subtraction, multiplication, and division.
# Accept two numbers and an operator as input.
# Return the calculated result.
# Handle division by zero appropriately.
#
# Examples:
# calculator(10, 5, "+") -> 15
# calculator(10, 5, "-") -> 5
# calculator(10, 5, "*") -> 50
# calculator(10, 5, "/") -> 2.0
def calculator(num1, num2, operator):
    if operator == "+":
        return num1 + num2
    elif operator == "-":
        return num1 - num2
    elif operator == "*":
        return num1 * num2
    elif operator == "/":
        if num2 == 0:
            return "Error: Division by zero"
        return num1 / num2
    else:
```

```
        return "Invalid operator"


print(calculator(10, 5, "+"))
print(calculator(10, 5, "-"))
print(calculator(10, 5, "*"))
print(calculator(10, 5, "/"))
```



Output

```
15
5
50
2.0
```

Task 2

Prompt and Code

```
# Write a Python function named sort_student_marks().
# Accept a list of student marks as input.
# Sort the marks in descending order, from highest to lowes
t.
# Do not modify the original list.
# Return the sorted list.
# Marks must be between 0 and 100.
# Ignore invalid marks outside this range.
#
# Example:
# Input: [75, 92, 60, 88]
# Output: [92, 88, 75, 60]
```

```
def sort_student_marks(marks):
    valid_marks = [mark for mark in marks if 0 <= mark <= 1
00]
    return sorted(valid_marks, reverse=True)


marks = [75, 92, 60, 88, 105, -10]

print(sort_student_marks(marks))
```



Output

```
[92, 88, 75, 60]
```

Task 3

Prompt and Code

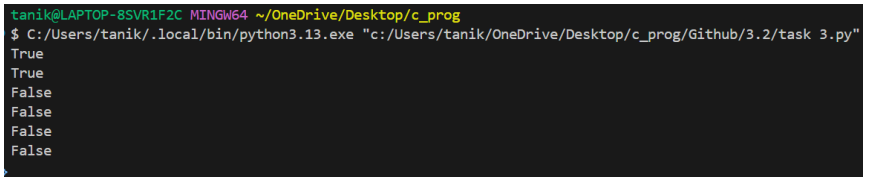
```
# Create a Python function named is_prime() that checks whe
ther a number is prime.
#
# Examples:
# Input: 2
# Output: True
#
# Input: 7
# Output: True
#
# Input: 10
# Output: False
#
# Input: 1
# Output: False
```

```
#
# Input: 0
# Output: False
#
# Input: -5
# Output: False
#
# A prime number is a positive integer greater than 1
# that has exactly two factors: 1 and itself.
# Return True for prime numbers and False otherwise.
def is_prime(number):
    if number <= 1:
        return False

    for i in range(2, int(number ** 0.5) + 1):
        if number % i == 0:
            return False

    return True


print(is_prime(2))
print(is_prime(7))
print(is_prime(10))
print(is_prime(1))
print(is_prime(0))
print(is_prime(-5))
```



Output

```
True
True
False
```

```
False
False
False
```

Task 4

Prompt

```
# Create a Python student grading system with a simple user
interface.
# Use Tkinter for the interface.
#
# The user should enter marks for 5 subjects.
# Each subject has a maximum of 100 marks.
# Calculate:
# 1. Total marks
# 2. Percentage
# 3. Grade
#
# Grade rules:
# 90-100: A
# 80-89: B
# 70-79: C
# 60-69: D
# Below 60: F
#
# Display the total marks, percentage, and grade clearly.
# Validate that marks are between 0 and 100.
```

Code

```
def calculate_grade():
    print("Student Grading System")
    print("-" * 30)
```

```
    marks = []

    for i in range(5):
        while True:
            try:
                mark = float(input(f"Enter marks for Subjec
t {i + 1}: "))

                if 0 <= mark <= 100:
                    marks.append(mark)
                    break
                else:
                    print("Marks must be between 0 and 10
0.")

            except ValueError:
                print("Please enter a valid number.")

    total = sum(marks)
    percentage = total / 5

    if percentage >= 90:
        grade = "A"
    elif percentage >= 80:
        grade = "B"
    elif percentage >= 70:
        grade = "C"
    elif percentage >= 60:
        grade = "D"
    else:
        grade = "F"

    print("\n" + "-" * 30)
    print("Result")
    print("-" * 30)
    print(f"Total Marks : {total:.0f}")
    print(f"Percentage  : {percentage:.2f}%")
    print(f"Grade       : {grade}")
```

```
calculate_grade()
```

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ C:/Users/tanik/.local/bin/python3.13.exe "c:/Users/tanik/OneDrive/Desktop/c_prog/Github/3.2/task 4.py"
Student Grading System
------------------------------
Enter marks for Subject 1: 90
Enter marks for Subject 2: 96
Enter marks for Subject 3: 95
Enter marks for Subject 4: 89
Enter marks for Subject 5: 97

------------------------------
Result
------------------------------
Total Marks : 467
Percentage  : 93.40%
Grade       : A
```

Output

```
Total Marks: 467
Percentage: 93.40%
Grade: A
```

Task 5

Prompt and Code

```
# Create a Python program for unit conversion between kilom
eters and miles.
#
# Create two functions:
# 1. kilometers_to_miles()
# 2. miles_to_kilometers()
#
# Use these conversion formulas:
# 1 kilometer = 0.621371 miles
# 1 mile = 1.60934 kilometers
#
# Return the result rounded to two decimal places.
# Handle negative values by displaying an appropriate error
```

```
message.
#
# Examples:
# kilometers_to_miles(10) -> 6.21
# miles_to_kilometers(10) -> 16.09
def kilometers_to_miles(km):
    if km < 0:
        return "Error: Distance cannot be negative"

    return round(km * 0.621371, 2)


def miles_to_kilometers(miles):
    if miles < 0:
        return "Error: Distance cannot be negative"

    return round(miles * 1.60934, 2)


print("10 km =", kilometers_to_miles(10), "miles")
print("10 miles =", miles_to_kilometers(10), "km")
```



Output

```
10 km = 6.21 miles
10 miles = 16.09 km
```